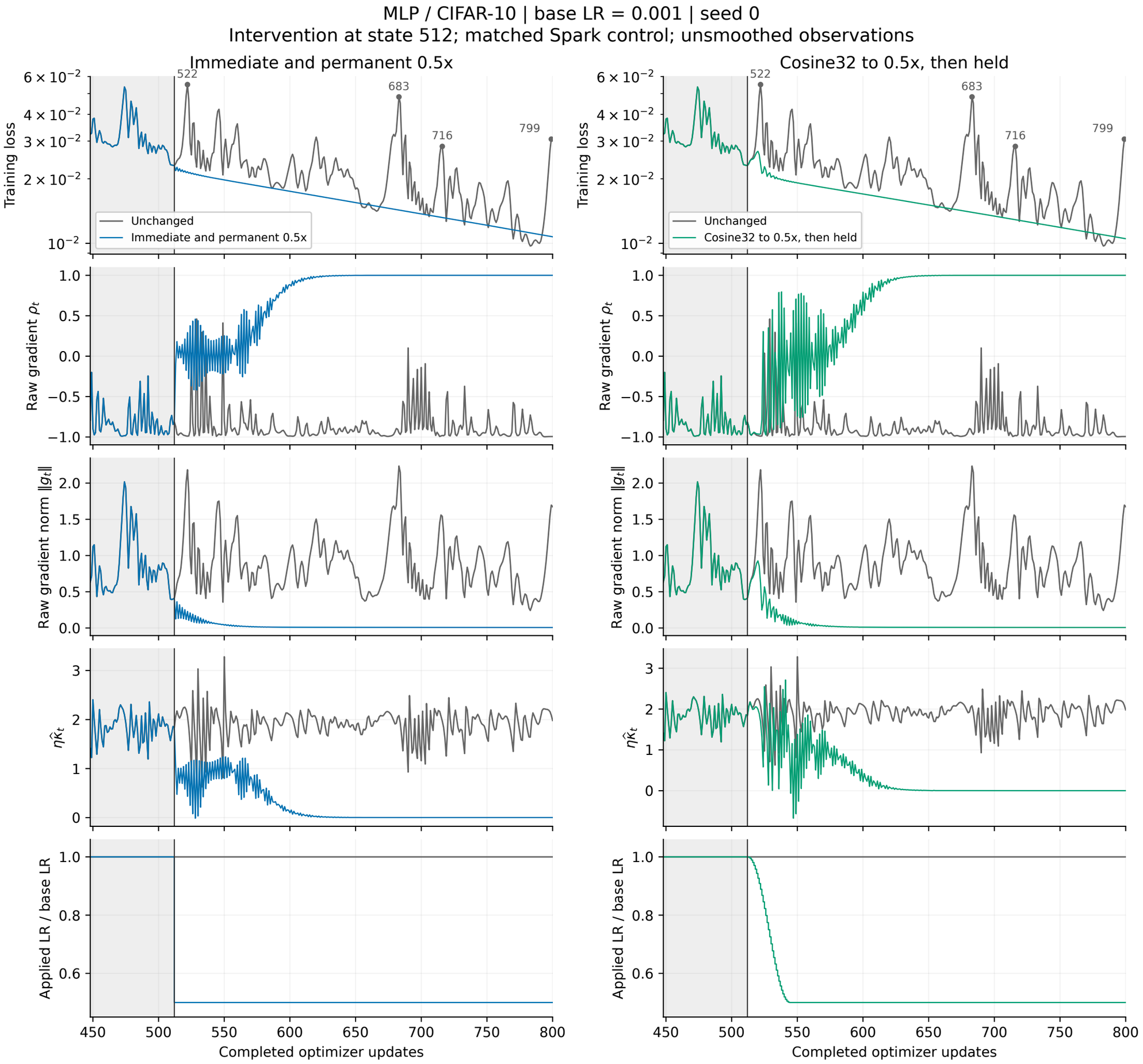

MLP / CIFAR-10 | base LR = 0.001 | seed 0
Intervention at state 512; matched Spark control; unsmoothed observations
Immediate and permanent 0.5x
Cosine32 to 0.5x, then held
Training loss
Unchanged
Immediate and permanent 0.5x
Cosine32 to 0.5x, then held
522
683
716
799
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\eta\hat{\kappa}_t$
Applied LR / base LR
Completed optimizer updates